

Simulating biomolecules with Structure-based Models

RNA Dynamics: Going from In Vitro to In Silico
March 19th, 2015

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Objective

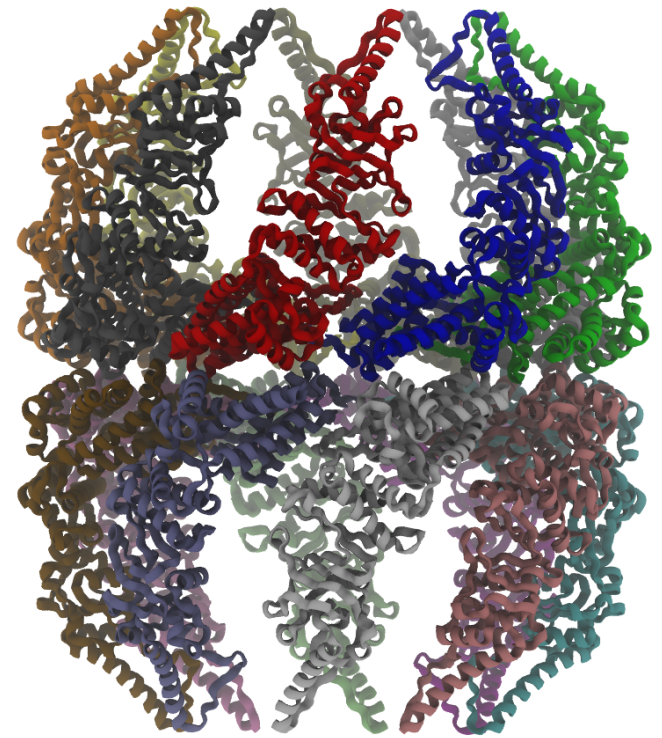
- *Give you a feel for what it means when someone says “we performed a simulation using a structure-based model”*
- **Today’s Exercises**
- Prepare and perform a C_α simulation with a Structure-Based Model (SBM)
 - Chaperonin
 - C_α model has $\sim 8,000$ atoms
 - All-atom model would have $\sim 60,000$ atoms
 - Observe local fluctuations
- Prepare and perform a simulation using an All-atom SBM
 - Ribozyme
 - All-atom model has ~ 7000 atoms
 - Observe local fluctuations

Overview of approach

- Download a PDB file
- Process using smog-server.org (Noel et al. *NAR* 2010)
 - i.e. make the model
- Perform the simulation using Gromacs
 - Integrate the equations of motion
- Visualize the simulation using VMD
 - Inspect the dynamics visually
- Analyze the simulated dynamics (won't do today)
 - Quantify the statistical/energetic characteristics

Case 1: Simulating a protein assembly with a coarse-grained model

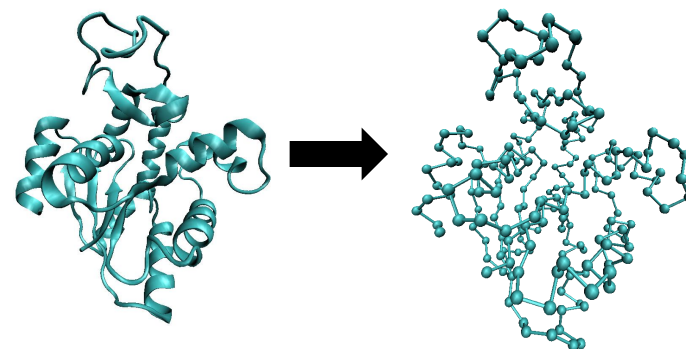
- Chaperonin
 - C_α model has $\sim 8,000$ atoms
 - All-atom model would have $\sim 60,000$ atoms
 - Characterize local fluctuations



C_α Structure-Based Model

Structure-based model

- A) C_α representation of backbone
- B) Only native interaction are stabilizing



bonds

angles

dihedrals

$$H = \sum_{bonds} K_r (r - r_0)^2 + \sum_{angles} K_\theta (\theta - \theta_0)^2 + \sum_{dihedrals} K_\phi^{(n)} [1 - \cos(n \times (\phi - \phi_0))]$$

$$+ \sum_{i < j-3} \left\{ \varepsilon(i, j) \left[5 \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - 6 \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{10} \right] + \varepsilon_2(i, j) \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} \right\}$$

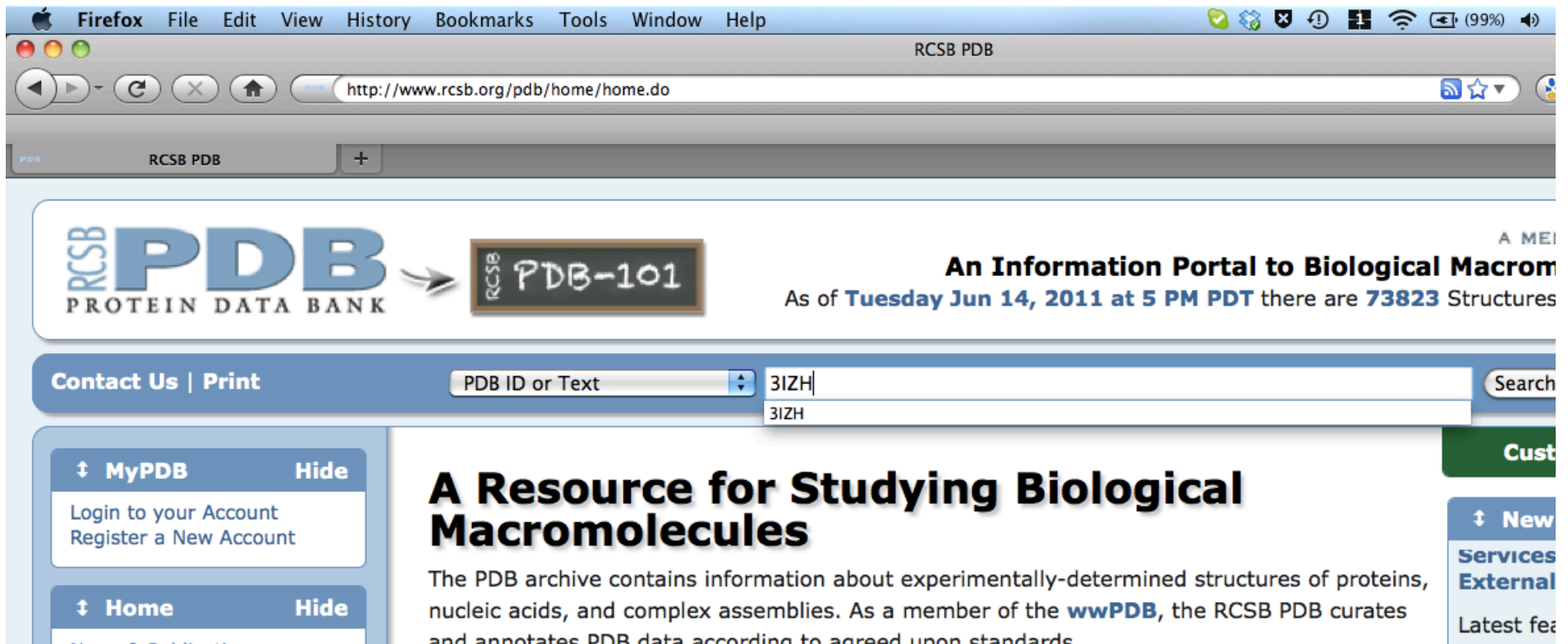
Shoemaker, Wang, & Wolynes, **PNAS**, 1997.
Nymeyer, Garcia & Onuchic, **PNAS**, 1995.

Contacts in native state: attractive

Non-contacts: repulsive

Download the Structure

- Normally: go to the PDB <http://www.rcsb.org>
 - Download pdb file 3IZH
- **To make your life easier, we provide the PDB at smog-server.org/rnasim-workshop/3IZH.pdb**



The screenshot shows a Firefox browser window displaying the RCSB PDB homepage. The address bar shows the URL <http://www.rcsb.org/pdb/home/home.do>. The page features the RCSB PDB logo, a search bar with the ID "3IZH" entered, and a sidebar with links to "MyPDB" and "Home". The main content area includes the heading "A Resource for Studying Biological Macromolecules" and a description of the PDB archive.

RCSB PDB
PROTEIN DATA BANK

PDB-101

An Information Portal to Biological Macromolecules
As of **Tuesday Jun 14, 2011 at 5 PM PDT** there are **73823** Structures

Contact Us | Print

PDB ID or Text:

MyPDB **Hide**
Login to your Account
Register a New Account

Home **Hide**

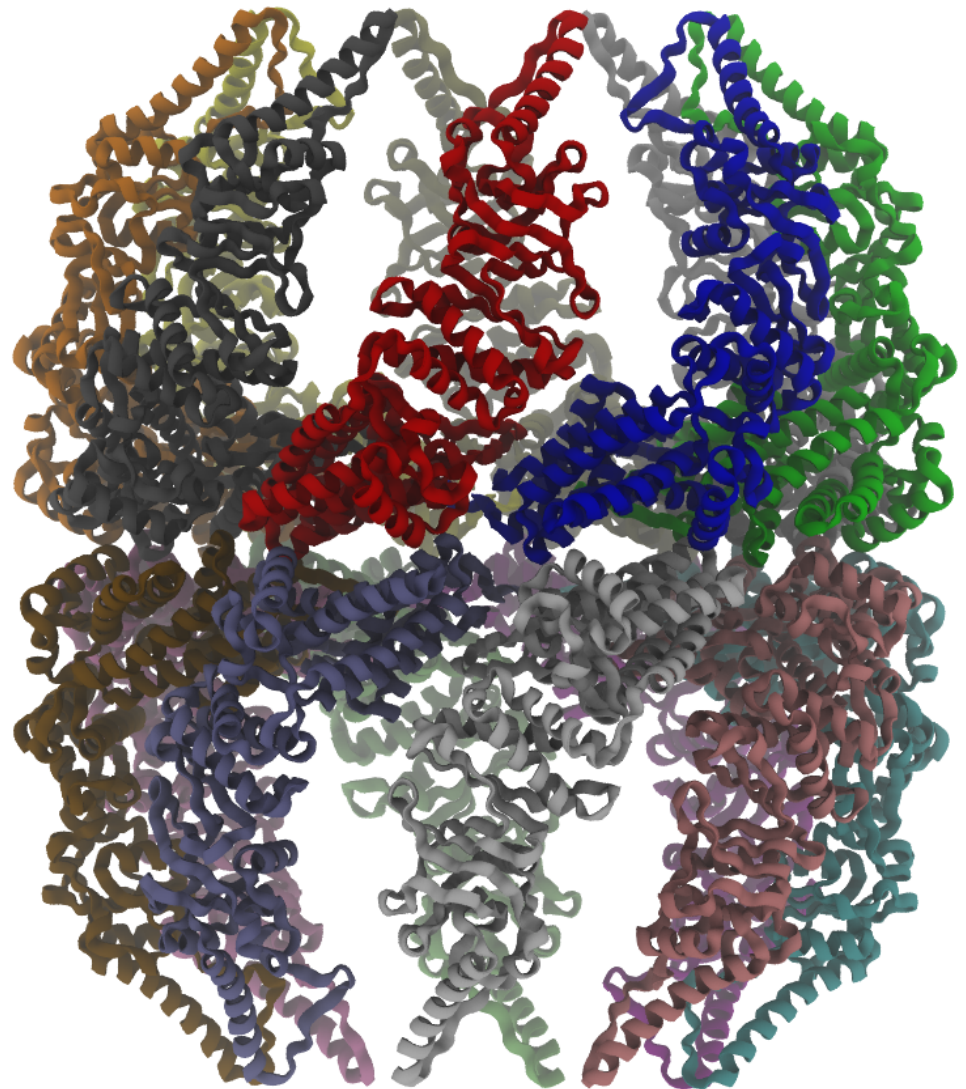
A Resource for Studying Biological Macromolecules
The PDB archive contains information about experimentally-determined structures of proteins, nucleic acids, and complex assemblies. As a member of the **wwPDB**, the RCSB PDB curates and annotates PDB data according to agreed upon standards

New Services External
Latest features

Visualize the system

View the pdb using VMD

From the command line, type
vmd 3IZH



Use the PDB file to generate input files using smog-server.org

- pdb: 3IZH
- If you are using a file from the PDB database, then make sure the pdb file conforms to the pdb guidelines for the webtool.
 - Only keep “ATOM”, “TER”, and “END”
- If you used our pre-prepared PDB file, then these details are taken care of.

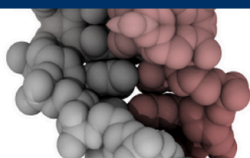
Go to smog-server.org



RICE



Northeastern



Introduction

Prepare a Simulation

Run a Simulation

Additional Tools

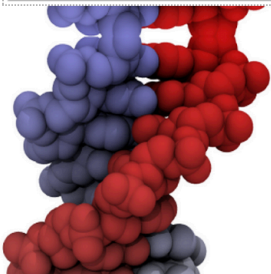
The Contact Map

Source Code Extensions

FAQs

News

Citing SMOG



SMOG: Structure-based Models for Biomolecules

A very brief introduction to structure-based modeling

While originally developed for the study of protein folding, over the last 10 years we have worked to extend the applicability of structure-based models to investigate a broad range of biomolecular dynamics, including domain rearrangements in proteins, folding and ligand binding in RNA and large-scale rearrangements in ribonucleoprotein assemblies. In its simplest form, a structure-based model defines a particular structure (normally obtained from X-ray, or NMR, methods) as the global energetic minimum. Oftentimes, this single-basin description can provide rich information about complex processes. When this simple representation is insufficient to account for the dynamics of interest, these models can be easily extended to include multiple basins of attraction, as well as describe non-specific effects (e.g. electrostatic or solvation effects). While coarse-grained variants of this model have had considerable success in expanding our understanding of biomolecular folding and function, in recent years, we have extended the Calpha model to an all-atom representation for structure-based simulations of biomolecules (complete descriptions can be found [elsewhere](#)). Additionally, with the computational simplicity of the models, they are proving to be very useful for structural modeling purposes, including atomic modeling of cryo-EM (e.g. [MDfit](#)) and [SAXS](#) data.

To increase the accessibility of these modeling techniques, we provide the smog-server webtool interface that will give you all the necessary information for performing structure-based simulations of proteins, nucleic acids, as well as many supported ligands. In addition, the webtool can handle all preprocessing steps that are necessary to prepare a structure-based forcefield for a given molecular structure. For in-depth descriptions of the underlying models, and for examples of how to use the models, we refer you to the [original publications](#) and recent [review articles](#).

How to prepare and simulate a SMOG model for a specific biomolecule

To run a single-basin structure-based model, the following steps should be adhered to.

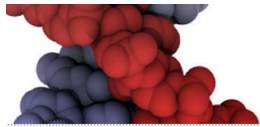
Step 0: Install the [Gromacs](#) molecular simulation package (Version 4.5.X). Gromacs will be the engine of the simulations. There are several excellent general [tutorials](#) available on the [Gromacs](#) webpage. We have found Gromacs to work very well for performing these simulations, both in serial and parallel, and it provides an excellent set of post-processing tools. However, to further extend the utility of these models, we are working to implement them in NAMD. Once that option is available, it will be announced here.

Step 1: [Prepare the structure-based potential input files for Gromacs](#). (All-atom or Calpha). Since this model is not included in any major software distribution, we have made it freely available through our [web interface](#). All you need to do is provide a pdb structure file and specify which parameters you would like to use, and we will generate the Gromacs input files for you.

Alternate Step 1: If you are familiar with the tool, try the beta version [SMOGv2.0.beta](#) that can be run locally on your machine. While we believe you will find it useful, it is still an experimental version and we appreciate feedback.

Step 2: [Run the simulation](#). Due to some technical differences between structure-based models and many other commonly used models, there are a few additional considerations that will ensure maximum performance of your calculations. Click [here](#) for a details on how to run a SMOG model in Gromacs. It is highly recommended that you read this information, even if you are an expert Gromacs

“Prepare a simulation”



Introduction

Prepare a Simulation

Run a Simulation

Additional Tools

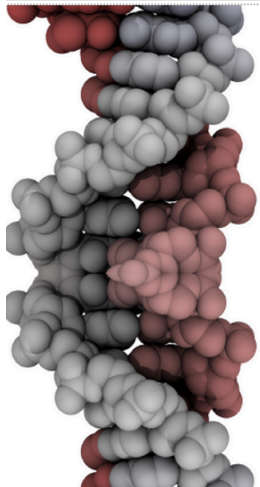
The Contact Map

Source Code Extensions

FAQs

News

Citing SMOG



Step 1: Generate Gromacs input files from a PDB structure

Version 1.2.2

[Version Notes](#)

Thank you for using the Noel-Onuchic-Whitford structure-based potential software. This module will take your pdb and contact file as input and produce a structure-based [Hamiltonian](#) with your specifications. The output TOP, GRO and index files can be used directly with [GROMACS](#), version 4.X.

Depending on your parameter selection, we kindly ask that you cite the applicable [primary references](#).

Fill in fields 1-12 and hit "Submit Query" in order to generate your input files for Gromacs

1. Upload PDB file 1GID.pdb (may be gzipped or b2zipped) [format requirements](#), [sample pdb/gro/top](#)

2. What contact map should be used? [Which map should I use?](#)

[Show Shadow Contact Map Options](#)

3. What level of graining would you like to use? [help](#)

4. Would you like to use the default forcefield, or customize the parameters? [what are default?](#)

5. How much empty space (in Å) would you like between the molecular system and the limits of the box? [Why is this important?](#)

X± Y± Z±

6. If you don't plan to use periodic boundary conditions, then you can check this box and your system will not be placed in a box (this will override the above spacing definitions) ☐

7. By default, the webtool will assign unit mass to each atom. If you would like to use heterogeneous masses that are roughly based on atomic masses (described [here](#)) check this box. ☐

8. By default, the webtool will assign zero charge to each atom. If you would like to include charges in your system (The exact charges included are described [here](#)) check this box. ☐

9. By default, if an atom name is not recognized by the webtool, the tool will stop and return an error message (Please [explain](#)). If you want the tool to ignore any unrecognized atoms (such as hydrogen) and continue to process the pdb, check this box. ☐

10. When using the All-atom model, you can use a longer timestep (0.002) if you increase the strength of the bond angles by a factor of 2 and weaken bonds by a factor of 2. To enable these modifications, check this box. ☐

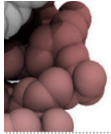
11. In what format do you want the output files? [Explain](#)

12. What nickname would you like to give this system?

After selecting the desired options, click "Submit Query". A link will be provided on the next page if the submission is successful.

Helpful Information

Page returned, download



Step 1: Generate Gromacs input files from a PDB structure

Version 1.2.2

[Version Notes](#)

Thank you for using the Noel-Onuchic-Whitford structure-based potential software. This module will take your pdb and contact file as input and produce a structure-based [Hamiltonian](#) with your specifications. The output TOP, GRO and index files can be used directly with [GROMACS](#), version 4.X.

Depending on your parameter selection, we kindly ask that you cite the applicable [primary references](#).

Generation of files completed. Read all output messages thoroughly, especially warning and error messages (EVEN IF YOU SUCCESSFULLY GENERATED YOUR FILES. There can still be important warnings in the .output file).

You can download the files [HERE](#).

If you are happy with your files, move on to [Step 2](#) and simulate your system.

DESCRIPTION OF OUTPUT FILES

Output messages

This field will provide you with lots of information about the processing. There is a wide variety of possible warning/error messages. It is **strongly suggested** that you read the warnings before using your files for simulations. Often warnings are not fatal, but you will be simulating something other than what you want. Here are a couple of common messages that may not be very transparent:

If you see this, great!

It means there were no errors.

Contents returned by page:

NAME = nickname you provided

NUMBER = assigned by webtool

- NAME.NUMBER.pdb – processed pdb file
- NAME.NUMBER.pdb.contacts – shadow map
- NAME.NUMBER.pdb.gro – CA gro file
- NAME.NUMBER.pdb.ndx – index file
- NAME.NUMBER.pdb.output –message file
- NAME.NUMBER.pdb.settings.used – settings
- NAME.NUMBER.pdb.top – CA top file

Contents returned by page:

NAME = nickname you provided

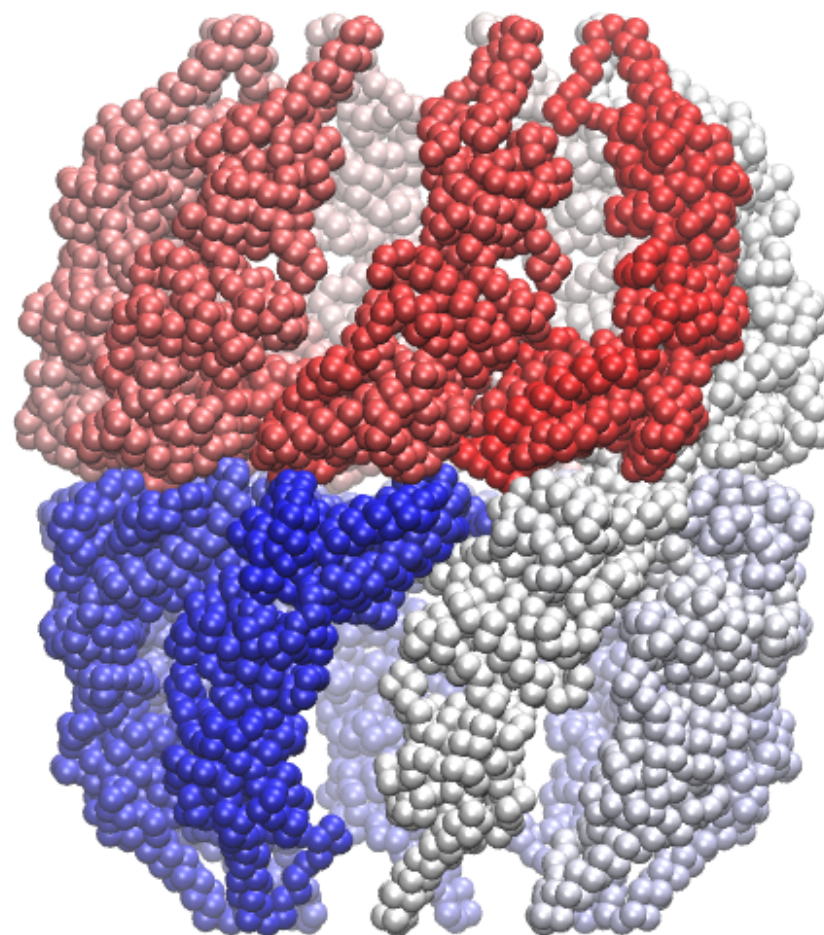
NUMBER = assigned by webtool

- **NAME.NUMBER.pdb – processed pdb file**
- NAME.NUMBER.pdb.contacts – shadow map
- **NAME.NUMBER.pdb.gro – CA gro file**
- NAME.NUMBER.pdb.ndx – index file
- NAME.NUMBER.pdb.output –message file
- NAME.NUMBER.pdb.settings.used – settings
- **NAME.NUMBER.pdb.top – CA top file**

gro (coordinate) file

Structure-Based gro file for Gromacs: smog@ctbp version 1.0.6

```
8208
 1 VAL  CA    1   9.734   5.862  11.552
 2 LEU  CA    2  10.018   6.106  11.622
 3 PRO  CA    3   9.975   6.483  11.668
 4 GLU  CA    4  10.047   6.668  11.338
 5 ASN  CA    5  10.400   6.817  11.388
 6 MET  CA    6  10.556   6.498  11.530
 7 LYS  CA    7  10.825   6.346  11.309
 8 ARG  CA    8  10.928   5.981  11.295
 9 TYR  CA    9  11.208   5.779  11.136
10 MET  CA   10  11.218   5.399  11.142
11 GLY  CA   11  11.424   5.100  11.025
12 ARG  CA   12  11.733   5.186  10.819
13 ASP  CA   13  11.638   5.558  10.818
14 ALA  CA   14  11.636   5.573  11.201
15 GLN  CA   15  11.972   5.393  11.200
16 ARG  CA   16  12.134   5.612  10.930
17 MET  CA   17  12.032   5.935  11.105
18 ASN  CA   18  12.140   5.825  11.454
19 ILE  CA   19  12.481   5.695  11.341
20 LEU  CA   20  12.555   6.002  11.131
21 ALA  CA   21  12.472   6.247  11.411
22 GLY  CA   22  12.691   6.052  11.658
23 ARG  CA   23  12.980   6.035  11.407
24 ILE  CA   24  12.985   6.413  11.343
25 ILE  CA   25  13.001   6.481  11.718
26 ALA  CA   26  13.287   6.228  11.742
27 GLU  CA   27  13.481   6.413  11.470
28 THR  CA   28  13.431   6.739  11.664
29 VAL  CA   29  13.577   6.603  11.991
30 ARG  CA   30  13.835   6.353  11.855
31 SER  CA   31  14.082   6.644  11.804
```



Visualize with “vmd name.gro”

.top (forcefield) file

Indicates that only CA atoms are in this simulation

Name of the complex

Specific atoms in this system

Contact definitions:

atom i, atom j, type, c10, c12

$$E = \frac{c12}{r_{i,j}^{12}} - \frac{c10}{r_{i,j}^{10}}, \quad c10 = 6\sigma^{10}, \quad c12 = 5\sigma^{12}$$

CA-CA backbone "bonds"

CA-CA-CA backbone bond angles

CA-CA-CA-CA backbone dihedral angles

```

Structure-based topology file for Gromacs: smog@ctbp version 1,0,6
[ defaults ]
;nbfunc comb-rule gen-pairs
      1          1 no

[ atomtypes ]
;name  mass      charge  ptype c6      c12
CA     1.000     0.000 A   0.000 0.16777E-04

[ moleculetype ]
;name  nrexcl
Macromolecule      3

[ atoms ]
;nr  type  resnr residue atom  cgnr charge  mass
  1  CA     1  VAL  CA     1  0.0   1.000
  2  CA     2  LEU  CA     2  0.0   1.000
...
...

[ pairs ]
  1  578 1  0.30873E-02 0.56583E-03
  1  579 1  0.87833E-02 0.19842E-02
...
...

[ bonds ]
;ai  aj  func  r0(nm)  Kb
  1  2  1  0.38091E+00 0.20000E+05
  2  3  1  0.38222E+00 0.20000E+05
  3  4  1  0.38511E+00 0.20000E+05
...
...

[ exclusions ]
;ai  aj
  1  578
  1  579
  1  582
...
...

[ angles ]
;ai  aj  ak  func  th0(deg)  Ka
  1  2  3  1  0.12475E+03 0.40000E+02
  2  3  4  1  0.11047E+03 0.40000E+02
  3  4  5  1  0.10419E+03 0.40000E+02
...
...

[ dihedrals ]
;ai  aj  ak  al  func  phi0(deg)  Kd(kJ/mol)  mult
  1  2  3  4  1  0.44326E+03 0.10000E+01 1
  1  2  3  4  1  0.13298E+04 0.50000E+00 3
  2  3  4  5  1  0.43680E+03 0.10000E+01 1
...
...

[ system ]
;name
Macromolecule

[ molecules ]

```

Setting up the run

- Download a .mdp file from
smog-server.org/rnasim-workshop
Filename CA.mdp

CA.mdp file (settings file)

```
;Run control: Use Langevin dynamics.
integrator = sd
;time step in reduced units. Since we use mass of a carbon=1, the time step is smaller than many other MD simulations.
dt = 0.0005
;number of integration steps
nsteps = 10000
;frequency to write coordinates to output trajectory .trr file
nstxout = 100000
;frequency to write velocities to output trajectory .trr file
nstvout = 100000
;frequency to write energies to log file
nstlog = 100
;frequency to write energies to energy file
nstenergy = 100
;frequency to write .xtc trajectory
nstxtcout = 100
;group(s) to write .xtc trajectory.
xtc_grps = system
;group(s) to write to energy file
energygrps = system
;Frequency to update the neighbor list (and the long-range forces,
;when using twin-range cut-off's).
nstlist = 20
;Make a grid in the box and only check atoms in neighboring grid cells
;when constructing a new neighbor list every nstlist steps.
ns_type = grid
;cut-off distance for the short-range neighbor list (NOTE: cut-offs should be longer if running a CA model than an AA model). 3.0 nm is about 2.5 times the typical longest interaction in a CA model. Using a cutoff of 2.0 nm is completely reasonable as a 12Å contact will have only ~1% of its potential energy at that distance. We leave 3.0 nm here for safety, but if you know what you are doing feel free to decrease to 2.0 nm. If you have longer contacts, then of course increase these numbers.
rlist = 3.0
;treatment of electrostatic interactions
rcoulomb = 3.0
;treatment of van der waals interactions
rvdw = 3.0
;this tells gromacs to use the tables that we give it, instead of the standard interaction types
coulombtype = User
vdwtype = User
; Periodic boundary conditions in all the directions
pbc = xyz
table-extension = 3 ; (nm) !IMPORTANT! Set this so that rvdw + table-extension equals half of the box's longest diagonal.
;Temperature coupling
tc-grps = system
tau_t = 1.0 ; Temperature coupling time constant. Smaller values = stronger coupling.
ref_t = 50.0 ; In reduced units (see Gromacs Manual for explanation. We find many proteins fold around T=100-120)
;Pressure coupling
Pcoupl = no
;Velocity generation
gen_vel = yes
gen_temp = 50.0
gen_seed = 30
ld_seed = 130
comm_mode = linear ; center of mass velocity removal. Make sure you select what is proper for your simulation.
;NOTE on comm_mode: When using periodic boundary conditions, which improves the speed of neighbor searching, removing the angular velocity will lead to a warning at the grompp stage of simulations. So long as the box size is properly sized, such that your system does not cross the boundaries, then this warning is harmless and you can simply issue -maxwarn X, where X is the number of allowed warning messages.
```

Prepare a tpr file

The tpr file stores all information about the simulation.

```
;-) G R O M A C S (-:
Gravel Rubs Often Many Awfully Cauterized Sores

;-) VERSION 4.5.1 (-:

Written by Emile Apol, Rossen Apostolov, Herman J.C. Berendsen,
Aldert van Buuren, Párr Bjelkmar, Rudi van Drunen, Anton Feenstra,

...
...

;-) grompp (-:

Analysing residue names:
There are: 8208 Protein residues
Analysing Protein...
This run will generate roughly 5 Mb of data

Option      Filename  Type      Description
-----
-f          CA.mdp   Input, Opt.  grompp input file with MD parameters
-po         mdout.mdp  Output      grompp input file with MD parameters
-c          Chaperonin,3864.pdb,gro Input      Structure file: gro g96 pdb tpr
etc.
-r          conf.gro   Input, Opt.  Structure file: gro g96 pdb tpr etc.
-rb         conf.gro   Input, Opt.  Structure file: gro g96 pdb tpr etc.
-n          index.ndx   Input, Opt.  Index file
-p          Chaperonin,3864.pdb,top Input      Topology file
-pp         processed.top Output, Opt. Topology file
-o          run.tpr    Output      Run input file: tpr tpb tpa
-t          traj.trr   Input, Opt.  Full precision trajectory: trr trj cpt
-e          ener.edr   Input, Opt.  Energy file

Option      Type  Value  Description
-----
-[no]h      bool  no     Print help info and quit
...
...
...
Generated 1 of the 1 non-bonded parameter combinations
Excluding 3 bonded neighbours molecule type 'Macromolecule'
Velocities were taken from a Maxwell distribution at 50 K
Number of degrees of freedom in T-Coupling group System is 24621.00

There was 1 note

gcq#189: "Been There, Done It" (Beavis and Butthead)
..
```

`grompp -f CA.mdp -c gro_file.gro -p top_file.top -o run.tpr`

run the simulation

Download the 'table' file at

smog-server.org/rnasim-workshop/table1012.xvg

Run the simulation:

```
mdrun -s run.tpr -v -table table1012.xvg -tablep table1012.xvg -noddcheck
```

~ 3-5 minutes on a 4 core 2-3 Ghz processor

Visualize the trajectory

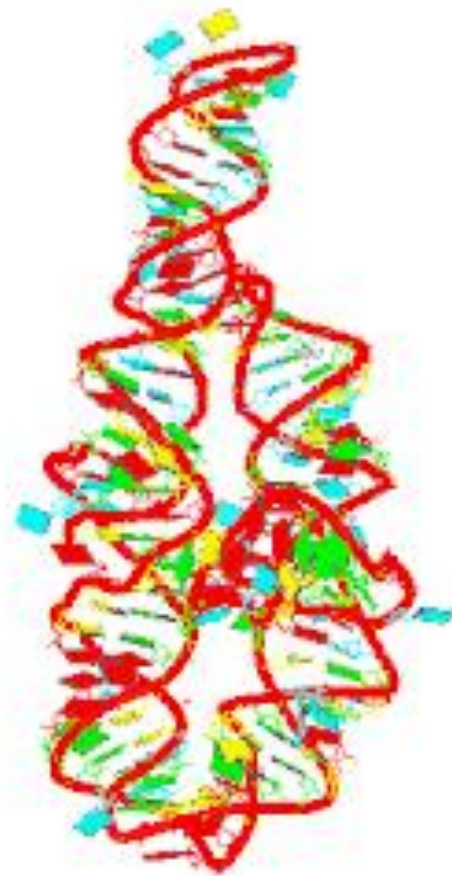
- From the command line, run

```
vmd grofile.gro traj.xtc
```

The trajectory should load automatically.

Case 2: Simulating an ribozyme using an all-atom structure-based model

- Group I ribozyme domain
 - 318 residues
 - AA model has ~7000 atoms
 - Again, characterize local fluctuations



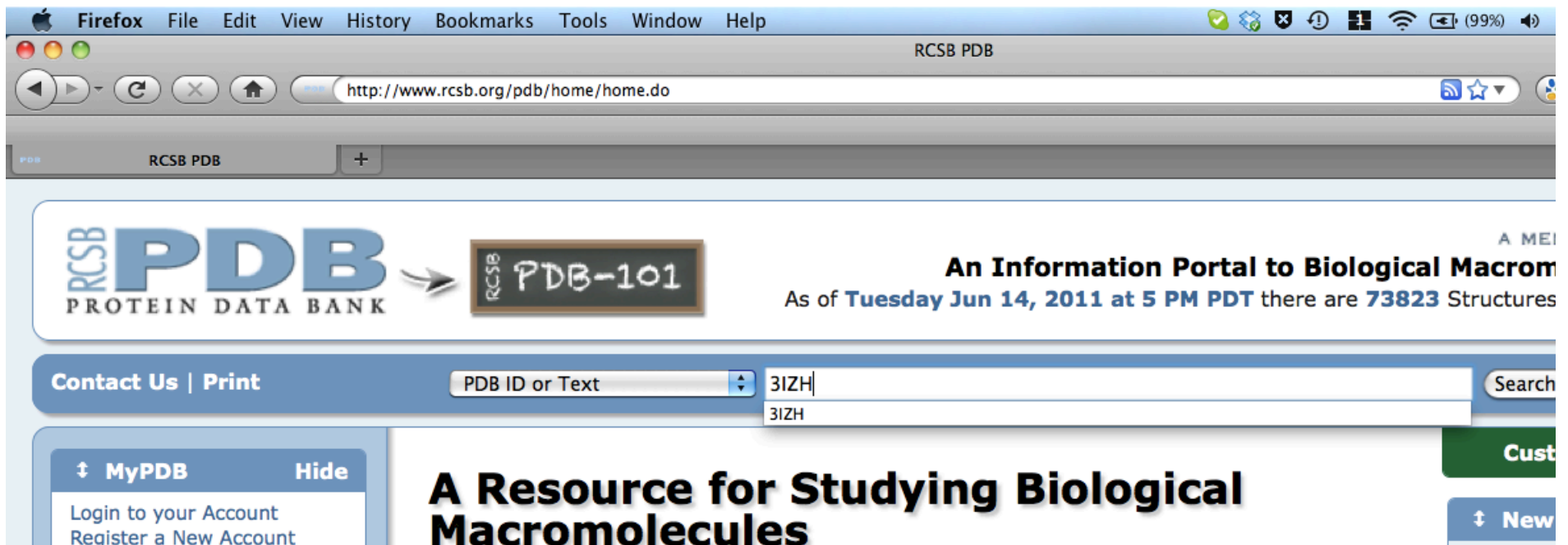
All-atom Structure-based model

- Energetically unfrustrated
 - no attractive interactions between non-native contacts
- All non-hydrogens included

$$E = \sum_{bonds} K_r (r - r_0)^2 + \sum_{angles} K_\theta (\theta - \theta_0)^2 + \sum_{dihedrals} K_{\phi, BB/SC}^{(n)} [1 - \cos(n \times (\phi - \phi_0))] \\ + \sum_{impropers/planars} (\chi_i - \chi_0)^2 + \sum_{i < j-3} \left\{ \varepsilon(i, j) \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - 2 \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \varepsilon_2(i, j) \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} \right\}$$

Download the Structure

- Go to the PDB <http://www.rcsb.org>
- Download pdb file **1GID**
- Or get it from:
smog-server.org/rnasim-workshop/1GID.pdb



Download the Structure

Or get it from: smog-server.org/rnasim-workshop/1GID.pdb

The screenshot shows the RCSB PDB website interface. The browser window is titled "RCSB PDB - 3IZH Structure Summary" and the address bar shows the URL "http://www.rcsb.org/pdb/explore/explore.do?structureId=3IZH". The page header includes the RCSB PDB logo, a "PDB-101" badge, and a banner stating "An Information Portal to Biological Macromolecular Structures". Below the header, there is a search bar with the text "PDB ID or Text" and a "Search" button. The main content area displays the structure summary for "Mm-cpn D386A with ATP" (3IZH). The summary includes the DOI "10.2210/pdb3izh/pdb", a primary citation from Cell (2011), and a list of authors. On the right side, there is a "Display Files" dropdown menu with options for downloading the structure in various formats: FASTA Sequence, PDB File (Text), PDB File (gz), mmCIF File, mmCIF File (gz), PDBML/XML File, PDBML/XML File (gz), and Biological Assembly (gz) (A). A 3D ribbon diagram of the protein structure is visible in the bottom right corner.

Visualize
the system

View the pdb using VMD

From the command line, type:

`vmd 1GID`

Use the PDB file to generate input files using smog-server.org

- pdb: 1GID
- Make sure the pdb file conforms to the pdb guidelines for the webtool.
 - Same steps as before, or use pre-prepared files

Three differences when running the All-atom model

- Download AA.mdp
 - This will be the settings file that we use for AA runs
- No table file needed this time since we use the built in 6-12 interactions
- Run the simulation (slightly easier command):
 - `mdrun -s run.tpr -v -noddcheck`
- Take a look when it works. Good luck!